Area Volume and Surface Area

Lesson 5 Polygons on the Coordinate Plane

Standards 6.G.A.3, 6.G.A.1

Learning target I can draw polygons from coordinates and use side lengths to find area.

Materials Rulers, grid paper, unit cubes or paper cube models, paper and scissors for nets.

Lesson Flow 50 Minutes

0–5 Activate prior knowledge. Read the learning target. Ask students what they already know about polygons on the coordinate plane. Use a quick model or estimate to make their starting ideas visible.

5–17 Teach and model. Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. Work through the two examples on the attached instruction page. Have students name the first step before showing it.

17–22 Guided practice. Revisit worked Example 2 with its solution covered. Students propose the next step and explain why it is valid; uncover and compare with the worked solution.

22–45 Independent classwork. Assign the attached 10 questions in order. Check items 1–3 for entry-level understanding, 4–6 for connected procedures, 7–8 for representation and explanation, and 9–10 for application or justification. Ask students to use the instruction-page notes area or extra grids for longer work.

45–50 Check understanding. Use classwork question 10 as an exit discussion: Plot and connect (-3,-2), (2,-2), (2,2), (-3,2). Find the side lengths and area. Find the perimeter too. Compare the result with the answer key and ask a student to defend the method.

Support and Extension

Support Model item 1 together with the listed tools, then have students attempt item 2 independently. Watch perpendicular height, missing faces, and square versus cubic units.

Extension Ask students to solve a final item another way, draw a second representation, or create a related problem with a checked answer.

Next step Record which students can explain the target independently. Begin the next teaching session with a brief correction for any recurring error.

Name _____ Date _____

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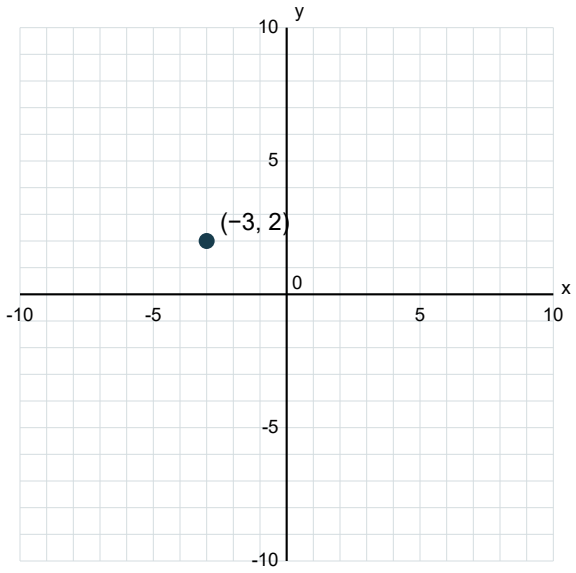
Learn and Discuss

Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths.

Example 1 Plot and connect (0,0), (8,0), (8,10), (0,10). Find the side lengths and area.
Solution Sides 8 and 10; area 80 square units. Horizontal length 8 - (0) = 8; vertical length 10 - (0) = 10; area = 8 × 10.
Example 2 Plot and connect (-5,-4), (4,-4), (4,4), (-5,4). Find the side lengths and area. Find the perimeter too.
Solution Sides 9 and 8; area 72 square units; perimeter 34 units. Horizontal length 4 - (-5) = 9; vertical length 4 - (-4) = 8; area = 9 × 8.

Classwork Show Your Work

Complete the 10 problems on the following page. Refer to the examples, draw models, and check your results.



Model a point (−3, 2) means move 3 left and 2 up from the origin. Label the axes and scale before plotting.

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Name _____ Show calculations, models, and units.

1. Plot and connect (0,0), (10,0), (10,12), (0,12). Find the side lengths and area.

2. Plot and connect (0,0), (11,0), (11,8), (0,8). Find the side lengths and area.

3. Plot and connect (0,0), (6,0), (6,10), (0,10). Find the side lengths and area.

4. Plot and connect (-2,-4), (2,-4), (2,2), (-2,2). Find the side lengths and area.

5. Plot and connect (-3,-7), (2,-7), (2,2), (-3,2). Find the side lengths and area.

6. Plot and connect (-8,-2), (3,-2), (3,3), (-8,3). Find the side lengths and area.

7. Plot and connect (-5,-7), (5,-7), (5,5), (-5,5). Find the side lengths and area. Find the perimeter too.

8. Plot and connect (-5,-4), (6,-4), (6,6), (-5,6). Find the side lengths and area. Find the perimeter too.

9. Plot and connect (-8,-5), (2,-5), (2,2), (-8,2). Find the side lengths and area. Find the perimeter too.

10. Plot and connect (-3,-2), (2,-2), (2,2), (-3,2). Find the side lengths and area. Find the perimeter too.

Polygons on the Coordinate Plane

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1. Sides 10 and 12; area 120 square units
Reasoning Horizontal length 10 - (0) = 10; vertical length 12 - (0) = 12; area = 10 × 12.

2. Sides 11 and 8; area 88 square units
Reasoning Horizontal length 11 - (0) = 11; vertical length 8 - (0) = 8; area = 11 × 8.

3. Sides 6 and 10; area 60 square units
Reasoning Horizontal length 6 - (0) = 6; vertical length 10 - (0) = 10; area = 6 × 10.

4. Sides 4 and 6; area 24 square units
Reasoning Horizontal length 2 - (-2) = 4; vertical length 2 - (-4) = 6; area = 4 × 6.

5. Sides 5 and 9; area 45 square units
Reasoning Horizontal length 2 - (-3) = 5; vertical length 2 - (-7) = 9; area = 5 × 9.

6. Sides 11 and 5; area 55 square units
Reasoning Horizontal length 3 - (-8) = 11; vertical length 3 - (-2) = 5; area = 11 × 5.

7. Sides 10 and 12; area 120 square units; perimeter 44 units
Reasoning Horizontal length 5 - (-5) = 10; vertical length 5 - (-7) = 12; area = 10 × 12.

8. Sides 11 and 10; area 110 square units; perimeter 42 units
Reasoning Horizontal length 6 - (-5) = 11; vertical length 6 - (-4) = 10; area = 11 × 10.

9. Sides 10 and 7; area 70 square units; perimeter 34 units
Reasoning Horizontal length 2 - (-8) = 10; vertical length 2 - (-5) = 7; area = 10 × 7.

10. Sides 5 and 4; area 20 square units; perimeter 18 units
Reasoning Horizontal length 2 - (-3) = 5; vertical length 2 - (-2) = 4; area = 5 × 4.